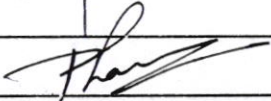


ENGINEERING CHANGE REQUEST

Customer: Graefelt	Project / Part Name: Suzuki J-0 L GEN2 / CH	Part number: 922447-0004e	Request date: 07/05/2025
1. Change Description: 1. Machining part number 922447-0002(G) 2. Assembly part number and rev 922448-0002(D) 3. N/A 4. HEX -128,129 (U slot aviable) 5. Hex no 31 $25^{\circ} \pm 0.1^{\circ} (2x)$ 6. Hex no 155 Position of M12 to datum K,C,D 7. Hex no 151 Flatness of N Plane 0.03 1.2 Process Modification: 1. Machining part number and rev changed to 922447-0004(C) 2. Assembly part number and rev changed to 922448-0004(B) 3. Page no 17: Hex no 173, 174, 175 was added (Compressor housing side 2.85 hole was added) 4. HEX -128,129 (U slot removed) 5. Hex no 31 $25^{\circ} \pm 10^{\circ} (2x)$ 6. Hex no 155 Position of M12 to datum K,C,A 7. Hex no 151 Flatness of N Plane 0.05 1.3 Modification reason: Customer design change before SOP			
Change priority: ☒ Emergency ☐ Urgent ☐ Routine		Required change date: 19/05/2025	
Projects affected:			
Change Assessment Record No: (Filled by Engg Section) MEC-505-25		Implementation Verified By / Date: CH. Malendra 01/07/25.	
Requested by: Ch. Paulson	Checked by: 	Approved by: R. Subramanian	

CHANGE ASSESSMENT RECORD

We are following "IATF 16949 CAPD Method 10.3 Continual Improvement Spirit" to improve our GTI.

Customer: Granett	Part Name: SUZUKI 1-0 LGFN2/CH	Part number: 922447-004	Change No.: MEC-505-25 Change date: 09/05/2025
--------------------------	---------------------------------------	--------------------------------	---

1. Change Description:

1.1 Original Process:

1. Machining part number 922447-0002(G)
2. Assembly part number and rev 922448-0002(D)
3. N/A
4. HEX -128,129 (U slot aviable)
5. Hex no 31 25*±0.1*(2x)
6. Hex no 155 Position of M12 to datum K,C,D
7. Hex no 151 Flatness of N Plane 0.03

1.2 Process Modification:

1. Machining part number and rev changed to 922447-0004(C)
2. Assembly part number and rev changed to 922448-0004(B)
3. Page no 17: Hex no 173, 174, 175 was added (Compressor housing side 2.85 hole was added)
4. HEX -128,129 (U slot removed)
5. Hex no 31 25*±10*(2x)
6. Hex no 155 Position of M12 to datum K,C,A
7. Hex no 151 Flatness of N Plane 0.05

1.3 Modification reason:
Customer design change before SOP

2. Feasibility Verification:

2.1 New Equipment/Facilities/Gages required: ☒ Yes ☐ No If Yes: **need to add new gauge.**

2.2 Additional man power required: ☐ Yes ☒ No If Yes:

2.3 QCD target achievement: ☒ Yes ☐ No If No:

2.4 CFT Training required: ☐ Yes ☐ No If Yes: **Awareness about new process**

2.5 Possibility to Modify: ☒ Yes ☐ No If No:

Others (if any):

3. Risk Analysis:

3.1 Investment Required: ☒ Yes ☐ No If Yes: **need to purchase new gauge.**

3.2 Inventory Loss(Existing product): ☐ Yes ☒ No If Yes:

3.3 Process effect: ☒ Yes ☐ No If Yes: **u slot removed.**

3.4 Product effect: ☒ Yes ☐ No If Yes: **compressor housingside 2.85 hole added.**

3.5 Productivity Effect: ☐ Yes ☒ No If Yes:

3.6 Process modification effect HSE system?: ☐ Yes ☒ No If Yes:

Risk Level: ☐ Minor ☐ Medium ☒ Critical

Others (if any):

This change was initiated by customer. only.

4. Customer communication required?: ☒ Yes ☐ Not Applicable

5. Customer Response: ☒ Accepted ☐ Rejected ☐ Not Applicable

6. Need to Communicate to:

6.1. Machining Section: ☒ Yes ☐ Not Applicable	6.6. QA Dept: ☒ Yes ☐ Not Applicable
6.2. Tool Section: ☐ Yes ☐ Not Applicable	6.7. QA Calibration: ☒ Yes ☐ Not Applicable
6.3. Maintenance Section: ☐ Yes ☐ Not Applicable	6.8. Business Dept: ☐ Yes ☒ Not Applicable
6.4. Post Machining section: ☒ Yes ☐ Not Applicable	6.9. HSE Dept: ☐ Yes ☒ Not Applicable
6.5. QC Section: ☒ Yes ☐ Not Applicable	7.0. Supplier: ☐ Yes ☒ Not Applicable

8. Trial Results: Start Date: **09/05/2025** End Date: **09/05/2025**
Trial performed ok

Requirements		Assessment		Responsibility	Completion date	Remarks
		Yes	No			
9.1 Add or modify fixture	Fixture Drawing	✓				
	Periodic Fixture Check Record	✓		Mahendra	15/05/2025	
9.2 Add or modify tool	Tool Drawing		✓			
	Cutting Tool List	✓		Deepak	11/06/2025	
9.3 Add or modify Inspection Method	Gage drawing	✓		Pinesh	11/06/2025	Calibrated By/Date:
	Gage List	✓		Anil	11/06/2025	
	CMM program	✓		Muni	11/06/2025	
	CNC Program	✓		Pandurangh	11/06/2025	
9.4 Add or modify new process	Process Flow Chart	✓		Mahendra	23/07/2025	
	Control Plan	✓		Mahendra	23/07/2025	
	Process Flow Chart	✓		Mahendra	23/07/2025	
9.5 Update PPAP documents	FMEA	✓		Mahendra	23/07/2025	
	Control Plan	✓		Mahendra	23/07/2025	
	Operator Inspection Sheet	✓		Mahendra	23/07/2025	
	Inspector Inspection Sheet	✓		Mahendra	23/07/2025	
	Control Plan attached drawing	✓		Mahendra	23/07/2025	
	KCC CIC List	✓		Mahendra	23/07/2025	
	EPCP	✓		Mahendra	23/07/2025	
	Appearance Criteria	✓		Mahendra	23/07/2025	
	Traceability Evidence	✓		Mahendra	23/07/2025	
	9.6 Add or modify Work instruction or Standard Operating procedure in Production Line	Casting Inspection WI	✓		Mallesh	
Machining Operation Instruction		✓		Mallesh	11/07/2025	
Loading & Unloading Procedure		✓		Mallesh	10/07/2025	
Machining Identification Instruction		✓		Mallesh		
Deburring WI		✓		Mallesh	10/07/2025	
Endoscope Inspection WI		✓		Mallesh	10/07/2025	
Ultrasonic Washing WI & SOP		✓		Mallesh	10/07/2025	
Position Gage Checking WI		✓		Mallesh		
End of line WI		✓		Mallesh	27/06/2025	
Assembly standard operating procedure.		✓		Mallesh	10/07/2025	
Welding Operation SOP			✓			
Welding Machining Parameters WI			✓			
Poka Yoke Manifestation		✓		Mallesh	29/07/2025	
Leak Testing And Engraving Instruction		✓		Mallesh	29/07/2025	
Leak Testing Operation Procedure		✓		Mallesh	10/06/2025	
9.7 Update Add or modify Work instruction or Standard Operating procedure in Post Production	Jet Washing WI & SOP		✓			
	Drying Work Instruction & SOP		✓			
	Magnetic detection WI & SOP		✓			
	Final Inspection WI	✓		Mallesh	11/07/2025	
	Packing Work Instruction & SOP	✓		Mahendra	21/07/2025	
9.8 Update Checklists / Records	Fixture Spot Check List		✓			
	Machine Spot Check List		✓			
	Poka Yoke check sheet	✓		Mallesh	29/05/2025	
	Welding Parameters Check List		✓			
	MPI Check List		✓			
	Washing Check sheet		✓			
9.9 Update Aspect, impact and HIRA register			✓			
10. ERP Update		✓		Pavithra	15/05/2025	
Others:						
11. Implementation Date: 10/05/2025					12. Customer Approved: ☒ Yes ☐ No ☐ Not Applicable	
13. Before and After change comparison:						
13.1 Capability Study: N/A						
13.2 Rejection Rate (PPM): N/A						
13.3 Run@Rate achieved: N/A						
Others (If any):						
14. Existing product handling:						
Existing product has no effect						
Prepared: Ch. Deulast		Checked: [Signature]		Approved: R. Jashwanth		

RISK ANALYSIS

Customer	Ganveth	Project Name	SUZUKI 1-OL GEN	Change No	MEC-505-29
Part Name	Center housing	Part Number	922447-0004	Date	07/05/2025

Change Description: 1.Machining part number and rev changed to 922447-0004(C)
 2.Assembly part number and rev changed to 922448-0004(B)
 3.Page no 17:Hex no 173, 174, 175 was added (Compressor housing side 2.85 hole was added)
 4.HEX -128,129 (U slot removed)
 5.Hex no 31 25°±10°(2x)
 6.Hex no 155 Position of M12 to datum K,C,A
 7.Hex no 151 Flatness of N Plane 0.05

S.No.	Risk Description	Risk Level				Action Needed	Responsibility	Target Date	Remarks
		NA	Low	Medium	High				
1	Availability of Operators	✓							
2	Quality History & Performance	✓							
3	Skill Level / Training			✓		Awareness about new design	perchu		
4	Resource requirements	✓							
5	Investments			✓		Need to purchase new gauge	Anil		
6	Existing Stock availability	✓							
7	Development Cost	✓							
8	Manufacturing Technology	✓							
9	Fixture & Gauges requirement			✓		New gauge design in placement	prakash Anil		
10	Any New Machine/Technology	✓							
11	Cycle Time								
12	Product/Process complexity								
13	Process capability	✓							

Conclusion:

☐ Risks Known and acceptable

☒ Risks Known and acceptable with identified actions.

☐ Risks known and not acceptable

Risk Level:

☐ Minor

☒ Medium

☒ Critical

Sign-Off:

.....
 Team member / Title / Date
 (Mn-Technical)

.....
 Team member / Title / Date
 (Mn-Production)

.....
 Team member / Title / Date
 (Mn-QC)

.....
 Team member / Title / Date

.....
 Team member / Title / Date

.....
 Team member / Title / Date
 (Q.A)

Prepared by
 Ch. Prakash

Reviewed By
 [Signature]

Approved By
 R. Prakash

CHANGE NOTICE FORM

CN No:25-2025

Customer Information				Process/Product/Design change ("x" all applicable)			
Date:	07/05/2025			1	X	Design Change	
Customer Name:	Garrett			2		Die/Mold Change	
Project / Part Name:	Suzuki 1.0L Gen2 / CH			3		Test/Inspection Method Change	
Casting Part number old / Dwg rev / Part rev	922446-0002 / G / D			4		Jig/Tool Change	
Casting Part number new / Dwg rev / Part rev	922446-0002 / N / E			5		Machine Change	
Machining Part number old / Dwg rev / Part rev	922447-0002 / G / G			6		Manufacturing Method Change	
Machining Part number new / Dwg rev / Part rev	922447-0004 / N / C			7		Manufacturing Process Change	
Assembly Part number old / Dwg rev / Part rev	922448-0002 / D / D			8		Material Change	
Assembly Part number new / Dwg rev / Part rev	922448-0004 / H / B			9		Sub-supplier Change	
Sub-component old Part number new / Dwg rev / Part rev				10		Drawing change	
Sub-component new Part number new / Dwg rev / Part rev				11		Packaging changes	
ERP code				12		New production layout	
				13		Manufacturing Site Change	
Description	Old	New		14		Others	
Casting	PMHW244602						
Machining	PMHW244702						
Assembly	PAHW244802						
Sub-component							
Others							
Change description							
What is the proposed change?				Customer updated the drawings.			
What is the reason?				Drawings updated to the production release.			
Current (show picture or sketch)				Proposed Change (show picture or sketch)			
922447-0002 G 922446-0002 D 922448-0002 D Garrett HOUSING CENTER 922447 G				922447-0004 C 922446-0002 E 922448-0004 B Garrett HOUSING CENTER ASSEMBLY 922447 N			
Garrett HOUSING CENTER 922447 G				Garrett HOUSING CENTER ASSEMBLY 922447 N			
Distribution List				PPAP Requirement			
☒ QA ☒ Foundry QC ☒ Foundry Technical ☐ Foundry Production ☒ Machining QC ☒ Machining Technical				☐ Machining Production ☐ Others (Specify): ☐ PPAP Level 1 ☐ PPAP Level 2 ☐ PPAP Level 3 ☐ PPAP Level 4 ☒ PPAP Level 5 ☐ Others			
Additional or other requirements:							
Stock Details by PMC and MSPMC				Signature		Date	
Castings Machined Parts Assy Parts In Transit Warehouse stock				Sub components: NA		Returnable packing: NA	
Evaluation				Resp.	Signature	Y	N
Is change feasible				Technical			
Is change risk acceptable				Technical			
Are required documents (PPAP documents, Work instructions & Standards) updated				Technical			
Is tooling change possible				Technical			
Are quality requirements after change is acceptable				QA			
Are Quality documents updated				QA			
Is material handling method OK?				Business		x	
Is cost impact communicated with customer				Business		x	
Approvals (depending on the type of change additional approvals may be needed and will be added to the list)							
Department		Date	Approval (sign)		Comment		
Business		07/05/2025					
Business		07/05/2025					
Final Disposition of old Parts / Comments: New ERP codes not yet generated, once release shared through mail.							

EVALUATION REPORT

Feasibility analysis

Feasible to implement in current process

Risk Analysis

→ U slot deleted and hole was added

Documents Impact

No	Document name	Resp	Target date	Remarks
	PPAP documents	malendra	22/09/25	/
	production documents	chandra	22/09/25	

Tooling / Material / Process / packing Impact Details

→ need to purchase new gauges (plug & chamfer)

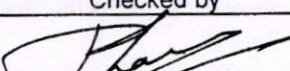
Cost Impact Details

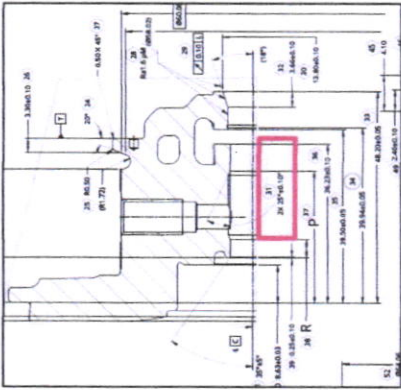
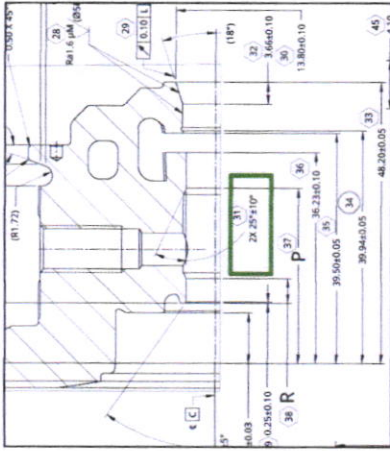
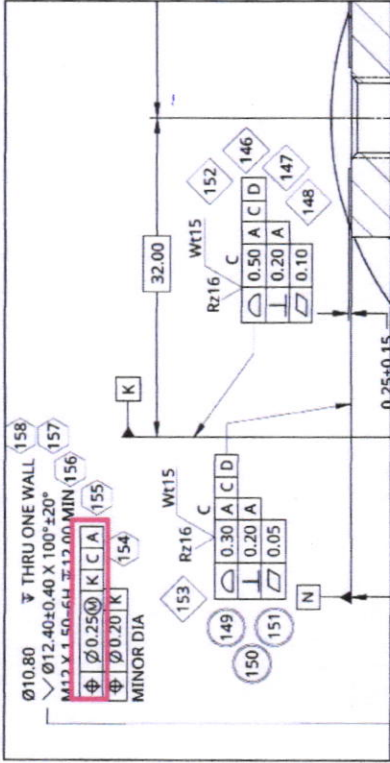
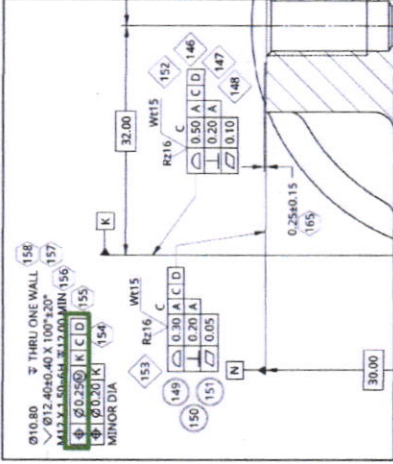
→ need to purchase new gauges

Traceability

Before		After	
--------	--	-------	--

Implementation status

Change start date		
Change implementation date		
Approved by	Checked by	Prepared by
R. Subramanian		CH. malendran

Machining Group			Drawing comparison		PAGE: 2 / 5
CUSTOMER:	Garrett	PROJECT/ PART NAME:	Suzuki 1.0L GEN2 / Center housing		
Before Change		After Change			
Machining: 922447-0002		Machining: 922447-0004			
					
					
HOD: R. Subramanian	QC: S. Sankar	Production/PPAP: S. V. S.	Technical: T. Manish S. K. May 100		

Machining Group		Drawing comparison		PAGE: 3 / 5																																																										
TITLE:		PROJECT/ PART NAME:																																																												
CUSTOMER:	Garrett	Suzuki 1.0L GEN2 / Center housing																																																												
Before Change		After Change																																																												
Machining: 922447-0002		Machining: 922447-0004																																																												
CAD DRAWING CHANGES TO THIS DRAWING MUST BE INCORPORATED VIA THE CAD SYSTEM <table><thead><tr><th colspan="4">REVISIONS</th></tr><tr><th>ZONE</th><th>REV</th><th>DECO/ECO</th><th>DESCRIPTION</th><th>MM/DD/YY</th></tr></thead><tbody><tr><td>SHT2</td><td>F10</td><td></td><td>-2, SYM 51,56,63,65,70 & 71 TOL WAS RUNOUT</td><td></td></tr><tr><td>SHT2</td><td>F11</td><td></td><td>-2, SYM 90,91 & 113 TOL WAS RUNOUT</td><td>07/25/24</td></tr><tr><td>TAB</td><td>G1</td><td>E2401501</td><td>922446-0002, -2, PRODUCTION RELEASED</td><td></td></tr><tr><td>S1C3</td><td>G2</td><td></td><td>-2, UPDATED NOTE 13</td><td>10/23/24</td></tr></tbody></table>		REVISIONS				ZONE	REV	DECO/ECO	DESCRIPTION	MM/DD/YY	SHT2	F10		-2, SYM 51,56,63,65,70 & 71 TOL WAS RUNOUT		SHT2	F11		-2, SYM 90,91 & 113 TOL WAS RUNOUT	07/25/24	TAB	G1	E2401501	922446-0002, -2, PRODUCTION RELEASED		S1C3	G2		-2, UPDATED NOTE 13	10/23/24	<table><thead><tr><th colspan="4">REVISIONS</th></tr><tr><th>ZONE</th><th>REV</th><th>DECO/ECO</th><th>DESCRIPTION</th><th>MM/DD/YY</th></tr></thead><tbody><tr><td>SHT2</td><td>F10</td><td></td><td>-2, SYM 51,56,63,65,70 & 71 TOL WAS RUNOUT</td><td></td></tr><tr><td>SHT2</td><td>F11</td><td></td><td>-2, SYM 90,91 & 113 TOL WAS RUNOUT</td><td>07/25/24</td></tr><tr><td>TAB</td><td>G1</td><td>E2401501</td><td>922446-0002, -2, PRODUCTION RELEASED</td><td></td></tr><tr><td>S1C3</td><td>G2</td><td></td><td>-2, UPDATED NOTE 13</td><td>10/23/24</td></tr></tbody></table>			REVISIONS				ZONE	REV	DECO/ECO	DESCRIPTION	MM/DD/YY	SHT2	F10		-2, SYM 51,56,63,65,70 & 71 TOL WAS RUNOUT		SHT2	F11		-2, SYM 90,91 & 113 TOL WAS RUNOUT	07/25/24	TAB	G1	E2401501	922446-0002, -2, PRODUCTION RELEASED		S1C3	G2		-2, UPDATED NOTE 13	10/23/24
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HOD: R. Subramanian	QC: Y. Subramanian	Production/PPAP: S. Subramanian	Technical: T. Malesh S. K. M. 1000																																																											
Greentech Industries (India) Pvt Ltd.																																																														

Machining Group		Drawing comparison		PAGE:	4 / 5
CUSTOMER:	Garrett	PROJECT/ PART NAME:	Suzuki 1.0L GEN2 / Center housing		
		Before Change	After Change		
		Machining: 922447-0002	Machining: 922447-0004		
HOD: R. Subramanian	QC: [Signature]	Production/PPAP: Sawdh	Technical: T. Maresh SK-MAN-VOC		

PAGE:

 $\frac{4}{5}$

Drawing comparison

Garrett

PROJECT/ PART NAME:

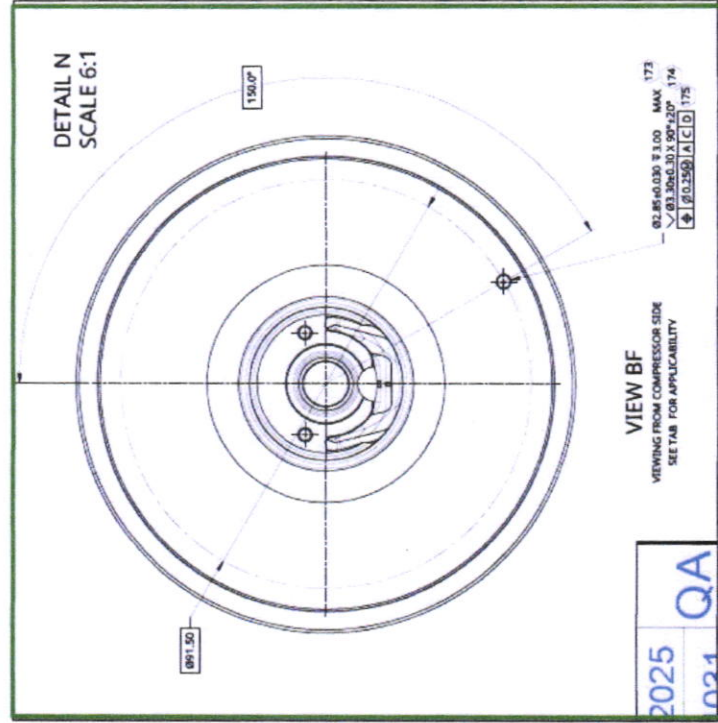
Suzuki 1.0L GEN2 / Center housing

Before Change

After Change

Machining: 922447-0002

Machining: 922447-0004



HOD:

QC:

Production/PPAP:

50

Technical:

Greentech Industries (India) Pvt Ltd.

Machining Group		Drawing comparison		PAGE: 5/5
TITLE:		PROJECT/ PART NAME: Suzuki 1.0L GEN2 / Center housing		
CUSTOMER: Garrett				
Before Change		After Change		
Machining: 922447-0002		Machining: 922447-0004		
HOD: <i>R. Subramanian</i>	QC: <i>Y. Subramanian</i>	Production/PPAP: <i>Savish</i>	Technical: <i>T. Malesh SK 7/11/2008</i>	
Greentech Industries (India) Pvt Ltd.				

Machining Group

TITLE:

Drawing comparison

PAGE:

1/2

CUSTOMER:

Garrett

PROJECT/ PART NAME:

Suzuki 1.0L GEN2 / Center housing

Before Change

Assembly:922448-0002

PART NUMBER	REV	CENTER HSG ITEM 1	COMPRESSOR INLET DIAMETER	PLUG ITEM 2	SEAL WASHER ITEM 3	PTN SPRAL ITEM 4	UNIQUE CALLOUTS	REMARKS	3D MODEL LINKED	RELEASE STATUS
922448-0002	D	922447-0002	G	434705-0002	444657-0003	756300-0010		SUZUKI 1.0L G PG	YES	PRODUCTION

CAD DRAWING

CHANGES TO THIS DRAWING MUST BE INCORPORATED VIA THE CAD SYSTEM

REVISIONS

ZONE	REV	DECO/ECO	DESCRIPTION	MM/DD/YY
-	-	D2309241	FILE RELEASED	04/24/24
-	A	D2400354	UPDATED 3D & 2D PER CTH CASTING 3D UPDATED	07/01/24
TAB B	B	D2403616	REVISED 3D & 2D	08/09/24
TAB C1	C1	D2405537	-1, 3D REVISED	10/23/24
TAB C2	C2		-1, PRE-PRODUCTION RELEASED	
TAB D	D	E2401501	-2, PRODUCTION RELEASED	

After Change

Assembly:922448-0004

PART NUMBER	REV	CENTER HSG ITEM 1	COMPRESSOR INLET DIAMETER	PLUG ITEM 2	SEAL WASHER ITEM 3	PTN SPRAL ITEM 4	UNIQUE CALLOUTS	REMARKS	3D MODEL LINKED	RELEASE STATUS
922448-0004	B	922447-0004	CN7381	434705-0002	444657-0003	756300-0010		SUZUKI 1.0L	YES	PRODUCTION

CAD DRAWING

CHANGES TO THIS DRAWING MUST BE INCORPORATED VIA THE CAD SYSTEM

REVISIONS

ZONE	REV	DECO/ECO	DESCRIPTION	MM/DD/YY
-	-	D2309241	FILE RELEASED	04/24/24
-	A	D2400354	UPDATED 3D & 2D PER CTH CASTING 3D UPDATED	07/01/24
TAB B	B	D2403616	REVISED 3D & 2D	08/09/24
TAB C1	C1	D2405537	-1, 3D REVISED	10/23/24
TAB C2	C2		-1, PRE-PRODUCTION RELEASED	
TAB D	D	E2401501	-2, PRODUCTION RELEASED	
TAB E	E	D2402707	-3, ADDED & FILE RELEASED	01/08/25
TAB F	F	D2500422	-4, ADDED & FILE RELEASED	01/22/25
TAB G	G	D2500425	-4, PRE PRODUCTION RELEASED	03/17/25
TAB H1	H1	E2500655	-2, OBSOLETE AND DELETED	
TAB H2	H2		DELETED ITEM 1 REV COLUMN	
TAB H3	H3		-4, PRODUCTION RELEASED	
TAB H4	H4		-4, 3D MODEL LINKED WAS NO	04/25/25

HOD:

R. Subramaniam

QC:

Signature

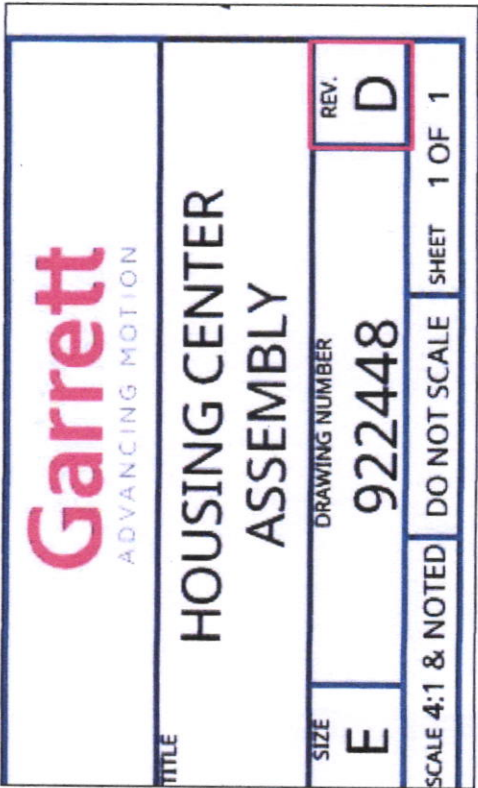
Production/PPAP:

Sprash

Technical:

T. Manish SK. Mayjood

Greentech Industries (India) Pvt Ltd.

Machining Group		Drawing comparison		PAGE: 2/2
CUSTOMER:	Garrett	PROJECT/PART NAME: Suzuki 1.0L GEN2 / Center housing		
Before Change		After Change		
Assembly:922448-0002		Assembly:922448-0004		
				
HOD: R. Subramanian	QC: S. Subramanian	Production/PPAP: S. Subramanian	Technical: T. Naresk SK. Naresk	